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Development and analysis of a Two stage Hybrid MPPT algorithm for solar PV systems

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Abstract

In this article, a hybrid Maximum Power Point Tracking (MPPT) approach that can quickly adapt to a fast changing weather conditions is proposed. The proposed method entails the combination of a Perturb and Observe (P&O) MPPT technique with a Look-Up Table (LUT) algorithm. The algorithm operates through a series of stages to attain the MPP with improved speed and accuracy, in response to the observed variations in the input(s). The efficacy of the proposed approach's MPP tracking performance is assessed through comprehensive simulation studies and experimental testing across diverse input conditions. The comparative evaluation of the proposed two-stage algorithm involves an assessment of its tracking response, reliability, and accuracy in relation to other established methodologies. The algorithm proposed exhibits a satisfactory level of overall efficiency, as evidenced by a tracking efficiency of 99.31% even under frequently changing weather conditions. This confirms the validity of the technique employed.
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Keywords: Photovoltaic (PV); Maximum Power Point Tracking; Perturb & Observe (P&O); Hybrid MPPT; Look Up Table

1. Introduction

The power generated by solar photovoltaic (SPV) system is significantly impacted by diverse environmental factors, particularly the cell temperature and solar irradiation, as evidenced by scholarly research [1]. The current output of a PV module exhibits a positive correlation with irradiation changes, while the voltage output of the PV system demonstrates an inverse relationship with temperature. Consequently, any alteration in temperature and/or

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irradiation results in deviation of the system from its peak operating point. Therefore, MPP trackers are integrated into an SPV system for extraction of maximum power. The basic requirements of a good MPPT controller are simplicity, rapid tracking under rapidly varying conditions, precision in MPP tracking, and less/no oscillation at MPP.

The MPPT algorithms are primarily categorized as true/direct search method, quasi-seeking/indirect search method, and hybrid method [1–4]. Some of the true-seeking MPPT algorithms are Perturb and Observe (P&O), Incremental Conductance (INC), Fractional Short Circuit Current (FSSC) and Open Circuit Voltage (FOCV) method [2,5]. Among the true-seeking algorithms, the P&O MPPT is a simple and popular approach [6,7]. However, this classical has a low tracking speed since it has a constant step size (usually 0.01). It suffers from power oscillation at MPP, and improper search during rapidly varying environmental conditions. Modified and improved versions of P&O MPPTs are proposed to improve the overall performance [6–9]. The Variable Step Size (VSS) and adaptive P&O algorithm works with different step values, for example, large (dD_{big}), and small (dD_{small}) step values. dD_{big} allows fast proximity to the MPP, and dD_{small} is applied to reach the MPP and less oscillation at the MPP. Such MPPT techniques possess improved tracking speed compared with the conventional P&O. However, these algorithms also suffer from MPP oscillation and poor trajectory in rapidly changing weather conditions [1,9]. The quasi-seeking method includes the computational intelligence-based MPPT and bio-inspired MPPT techniques. Fuzzy Logic Control (FLC), Particle Swarm Optimization (PSO), Look-Up Table (LUT) and Genetic Algorithm (GA) MPPT algorithms [10–15] are some of the quasi-seeking methods. The LUT method is considered the simplest method among quasi-seeking techniques. Unlike the other quasi-seeking techniques, the LUT method does not require any complicated computational algorithm, and training before implementation is not required. However, the LUT does not provide precise results under conditions not stated in the table and hence, this method is not very popular [1]. The third category of MPPT controllers is the hybrid MPPT algorithms, which are developed by combining more than one MPPT algorithm to improve the individual MPPT algorithms' overall performance and effectively utilize the combined advantages of multiple algorithms. In most cases, a hybrid MPPT is a combination one true-seeking and one quasi-seeking method. The INC and PSO hybrid MPPT technique is presented and demonstrates better tracking response and lower power oscillation [11]. FOCV and FSCC based hybrid MPPT are investigated in Moradi et al. [16], Sher et al. [17]. These algorithms work in two stages; in the first stage, the set point is computed using the measured value of V_{oc} or I_{sc}, and the MPP is tracked based on the calculated value. Then, to attain the MPP, P&O is used in the second stage. An MPPT using INC with Fractional Order (FO) optimization is proposed in Ammar et al. [18]. The fractional-order parameters are optimized using three different bio-inspired algorithms. A detailed simulation study is performed and evaluation of the performance of the PSO-optimized FO-INC PV system, Ant Colony Optimization (ACO) FO-InC system and Ant Lion Optimization (ALO) FO-INC PV system is performed. An improved hybrid MPPT algorithm using ANN and PSO method is proposed in Ibnelouad et al. [12]. This method combines a soft-computing technique with a bio-inspired approach for MPPT development. Elephant Herding Optimization (EHO) and Fuzzy Logic Controller-based hybrid MPPT algorithm proposed in Veeramanikandan and Selvaperumal [13] works in two stages.

The above discussions infer that the hybrid MPP Tracking techniques combine the conventional true-seeking methods like P&O, INC, FOCV, and FSCC techniques with computational intelligent techniques, meta-heuristic optimization approach, and bio-inspired techniques. The benefits and drawbacks of conventional true-seeking methods have been discussed in the literature. Computational intelligent and bioinspired MPPT algorithms offer quicker response and greater accuracy in most cases. However, such algorithms have a high computing load and, consequently, require expensive high-performance computing devices. In addition, specialized knowledge, extensive sensory gadgets, and large amounts of historical data are required to model, train, test, and develop MPPT systems. It makes the quasi-seeking methods complex. In a few cases, two quasi-seeking techniques are combined to develop hybrid MPPT algorithms. Although the performance achieved is comparatively better, the integration of two quasi-seeking techniques makes the system even more complex. This paper proposes a two-stage hybrid MPPT algorithm, which combines two simple MPPT techniques. The primary goal is to develop a simple, affordable, and easy to implement hybrid MPPT algorithm for PV systems with fast tracking response, fast tracking with better accuracy and minimal oscillation at the MPP.

2. Development of proposed MPPT algorithm

The proposed hybrid MPPT algorithm combines classical P&O and LUT methods. In the LUT, predefined datasets containing corresponding inputs and outputs are stored in memory. Data sets are typically generated based

on system knowledge or using available simulation/experimental data [1,14,15]. Instead of undergoing mathematical calculations, the LUT directly retrieves the output corresponding to the input from the preassigned data. In the proposed approach, Irradiance (G) and Temperature (T) are the inputs to the LUT, output is the duty cycle. The LUT method provides faster tracking because the output is extracted directly from the stored data. However, performance may be inaccurate under different environmental conditions. On the contrary, the P&O MPPT provides an accurate duty cycle under the cases where MPP is attained even though the tracking is slow. In addition to that the implementation of the selected techniques is straightforward and simple. Therefore, a hybrid MPPT using P&O and LUT that combines the advantages of both techniques is proposed and developed. When a large variation in the input is detected, the LUT provides a duty cycle based on the input(s). Further, P&O MPPT follows and tracks to reach MPP. Using this two-step approach, tracking speed is increased by LUT, and P&O MPPT increases tracking accuracy. The perturbation value of P&O MPPT is set as $dD = 0.005$ which is half of the classical P&O ($dD = 0.01$). As a result, the swing around the MPP is also reduced. As a result, the proposed method reaches the MPP quickly and the power swing around the MPP is comparatively less. The flowchart of the two-stage hybrid MPP tracking technique is illustrated in Fig. 1.

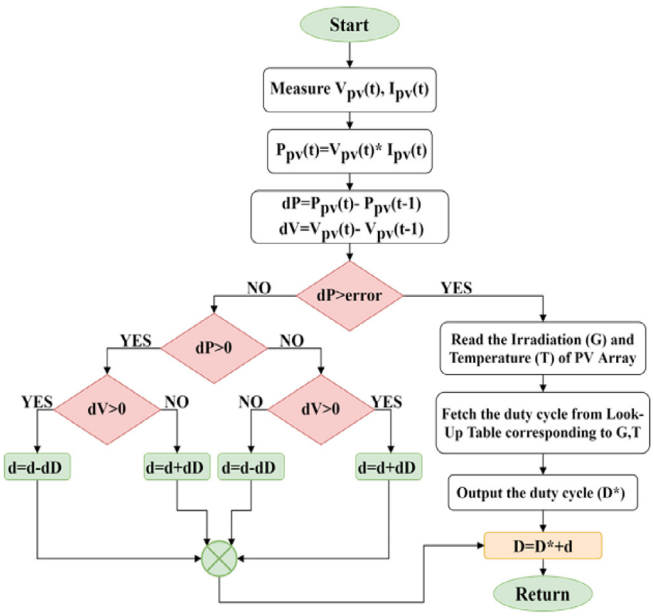


Fig. 1. Flowchart of the proposed MPPT.

Whenever there is a sudden change in PV input conditions, there will be a significant change of power. Under such circumstances, the value dP will become larger and the condition $dP>error$ will be satisfied. Under this circumstance, the algorithm will transition to a LUT and obtain a duty cycle from the LUT that corresponds to the input conditions. Subsequently, as the operating point approaches MPP using the designated duty cycle, the value of dP will decrease, resulting in the condition ($dP>error$) becoming false. When dP drops below the error value, the algorithm shifts to P&O and changes the duty cycle to reach MPP. Therefore, in the first stage, the duty cycle given by LUT will move the point of operating close to MPP very quickly. Then, in the second stage, P&O helps to attain the MPP in a few steps. As dD is assigned to 0.005 in the proposed algorithm, it minimizes the power swing at MPP. Optimizing the performance of the algorithm, specifically in relation to tracking speed with minimal deviation, involves tuning the error value through a trial-and-error approach. In the proposed system the error value is "$error=5$".

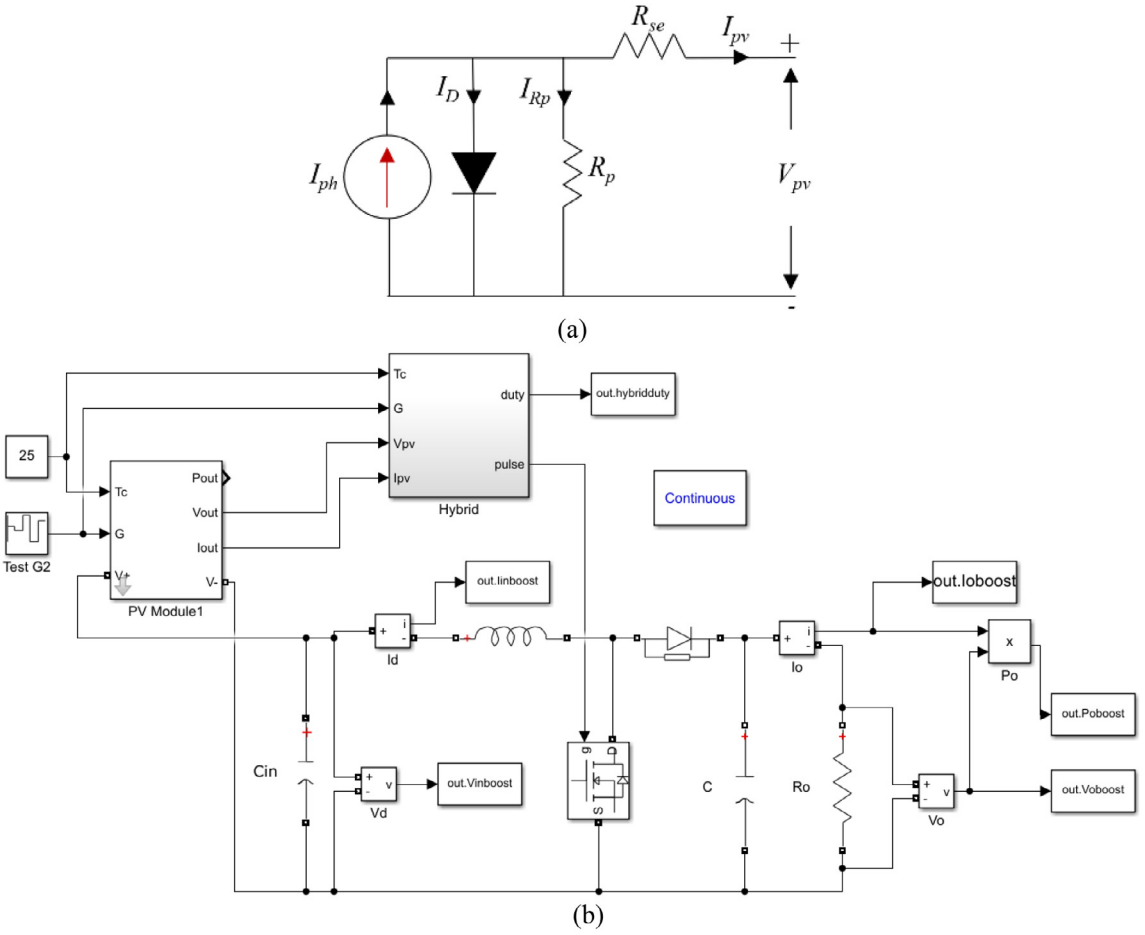


Fig. 2. (a) Solar PV cell model; (b) Simulation model of the solar PV system.

3. Modeling, and design of a PV system

Fig. 2(a) depicts the physical model of a solar PV cell. The current generated by a PV cell is given as.

$$I_{pv} = I_{ph} - I_s \left\{ \left(\frac{q}{AkT_cN_s} V + I R_{se} \right) - 1 \right\} - \frac{V + I R_{se}}{R_p} \tag{1}$$

As a result of its extremely high value, the parallel resistor (Rp) is excluded for study. Thus, the first Eq. (1) is restated as follows:

$$I_{pv} = I_{ph} - I_s \left\{ \left(\frac{q}{AkT_cN_s} V + I R_{se} \right) - 1 \right\} \tag{2}$$

Simulation studies were conducted using a solar PV module that is based on Eq. (2). The terms I_{ph}, I_s, and I_{pv} refer to the photo-generated current, reverse saturation current, and output current, respectively. T_c refers to the cell temperature, A is the ideality factor, and the Boltzmann constant is denoted by k. N_s represents the number of series connected cells in a PV module. In the proposed PV system, a DC–DC boost converter is used, and the standard design procedure outlined in Refs. [19,20] is used to calculate values of the converter components. Table 1 presents the specifications of the photovoltaic (PV) module, while Table 2 lists the values of the converter parameters and components. Fig. 2 depicts the physical model of a solar photovoltaic system, which has been modeled in MATLAB.

Table 1. PV Module specifications.

Description	Value
Peak Power (P_{max})	50 W_p
Current @ MPP	2.91 A
Voltage @ MPP	17.2 V

Table 2. Specifications of DC–DC Boost converter.

Description	Value
Input voltage (V_{in})	17 V
Output voltage (V_o)	24 V
Capacitors (C_{in}, C_{out})	100 μF, 50 μF
Inductor (L)	1.1 9 mH
Load Resistor (R_L)	11.54 Ω
Switching Frequency (f_{sw})	10 kHz

Table 3. Look Up Table.

Temp (°C)	Irradiation (W/m^2)									
	400	500	600	700	800	900	1000	1100	1200	1300
20	0.1	0.1	0.10	0.16	0.21	0.26	0.29	0.32	0.35	0.38
25	0.1	0.1	0.11	0.17	0.22	0.27	0.30	0.34	0.37	0.39
30	0.1	0.1	0.12	0.18	0.23	0.27	0.31	0.34	0.37	0.39
35	0.1	0.1	0.13	0.19	0.24	0.28	0.32	0.35	0.38	0.40
40	0.1	0.1	0.14	0.2	0.25	0.29	0.33	0.36	0.38	0.41
45	0.1	0.1	0.15	0.21	0.26	0.3	0.35	0.37	0.39	0.41
50	0.1	0.1	0.17	0.22	0.27	0.31	0.35	0.37	0.4	0.42
55	0.1	0.1	0.18	0.23	0.28	0.32	0.35	0.38	0.41	0.43

4. Simulation results and discussion

The performance evaluation of the proposed technique is conducted through simulation studies and the accuracy of duty cycle, tracking efficiency, and oscillation at MPP are assessed. The simulation studies are conducted using both slow changing and rapidly varying inputs. The overall performance is validated through a comparative analysis of its tracking performance against that of P&O and VSS Fuzzy MPPT algorithm.

To obtain input and output data for a LUT, simulation of the photovoltaic system utilizing a classical P&O algorithm is conducted across a diverse set of input parameters (T, G), with the resulting output in the form of duty cycle being recorded for each input condition. A LUT of dimensions 8×10 has been created based on the collected data. The LUT utilized for the proposed MPPT algorithm is presented in Table 3. The output (D^*) is generated by LUT in relation to the inputs. (T, G). An example scenario involves the utilization of the standard test conditions as inputs, resulting in an output of $D^* = 0.30$.

4.1. Simulation and analysis of PV system with proposed MPP tracking technique

The PV system inputs are varied at intervals of 100 ms, as presented in Table 4. The power output of PV system for the selected study cases with the two-stage hybrid MPPT algorithm is shown in Fig. 3.

Table 4. Input test conditions for Hybrid MPPT based PV system.

Study Case (SC)	SC 1 (0–50) ms	SC 2 (50–100) ms	SC 3 (100–150) ms	SC 4 (150–200) ms
Temperature (°C)	25	25	40	30
Irradiation (W/m^2)	1000	700	800	900

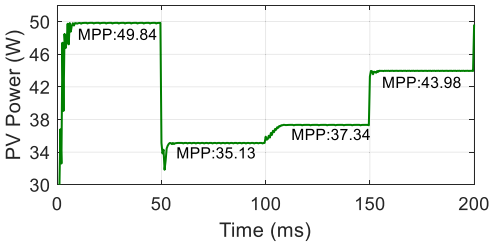


Fig. 3. PV output power with proposed Hybrid MPPT for the selected study cases.

Table 5. Power output and tracking efficiency of proposed Hybrid MPPT algorithm for selected study cases.

	Output Power (W)						
	Actual	P&O	η (%)	Fuzzy [2]	η (%)	Proposed	η (%)
Case 1	49.90	49.64	**99.40**	49.85	**99.89**	49.84	**99.87**
Case 2	35.38	34.47	**97.42**	35.21	**99.51**	35.13	**99.29**
Case 3	37.69	37.55	**99.62**	37.37	**99.15**	37.34	**99.07**
Case 4	44.42	43.89	**98.80**	44.00	**99.05**	43.98	**99.01**

Table 6. Step changing inputs.

Irradiation (W/m^2)	22	30	25	37	44	20
Temperature (°C)	950	1250	775	1175	875	900

The efficiency of PV system power output utilizing the proposed MPPT method is compared against the other MPPT methods, and the results are listed in Table 5. The overall tracking efficiency obtained for the selected study cases is 99.31%. The results confirm the accuracy of MPP tracking of the proposed MPPT algorithm.

Furthermore, step-changing inputs for temperature and irradiance are used to simulate the PV system. The default and out-of-range values are chosen in this instance as inputs. Table 6 provides the step-changing irradiation and temperature values. MPP tracking of Fuzzy MPPT, and the proposed method is shown in Fig. 4(a). In the suggested technique, the duty cycle that most closely resembles the input circumstances is pulled from the LUT at the first stage whenever a change in the input is observed. Following that, P&O MPPT tracks to reach the MPP in the second stage. The duty cycle output of the chosen algorithms and the proposed approach are depicted in Fig. 4(b). As can be shown, the suggested MPPT can precisely track the MPP under all the chosen circumstances.

As previously mentioned, the classical P&O algorithm experiences oscillations around the MPP upon continuous perturbations. This results in an elevation of the ripple magnitude in the converter output. It is possible to decrease the power swing by reducing the step value of duty cycle. Thus, the proposed approach sets the perturbation value of the P&O MPPT as $dD = 0.005$. Fig. 4(c) illustrates the duty cycle oscillation of proposed MPPT algorithms at MPP against the classical P&O method. The classical P&O exhibits an oscillation ranging from 0.29 to 0.31, with a discrete step size of 0.01. But, the MPPT under consideration exhibits oscillations within the range of 0.285 to 0.295, with a step increment of 0.005. The MPPT algorithm that exhibits a fuzzy logic approach displays a stable behavior at the MPP without any oscillatory tendencies. The hardware prerequisites for implementing fuzzy Maximum Power Point Tracking (MPPT) are considerably greater than those of the proposed technique, owing to the increased data requirements. In addition, it necessitates specialized knowledge of the system for the development. Fig. 4(d) illustrates the oscillation magnitude present in the PV output power while the system is operated with the P&O, Fuzzy, and Hybrid MPPT techniques. The proposed algorithm yields a lower output power ripple comparatively.

The tracking performance of the proposed tracking methodology in the presence of dynamic variations is evaluated by applying rapidly varying levels of irradiation to the PV system, as depicted in Fig. 5(a), while maintaining a constant temperature of 25 °C. Fig. 5(b) illustrates the duty cycle produced by the MPPTs that were examined. In scenarios where irradiation conditions are subject to rapid variation, the conventional P&O method for tracking is insufficient and results in increased oscillations during operation. The fuzzy MPPT demonstrates efficient tracking of MPP. But the accuracy and speed of tracking are very similar to the proposed algorithm.

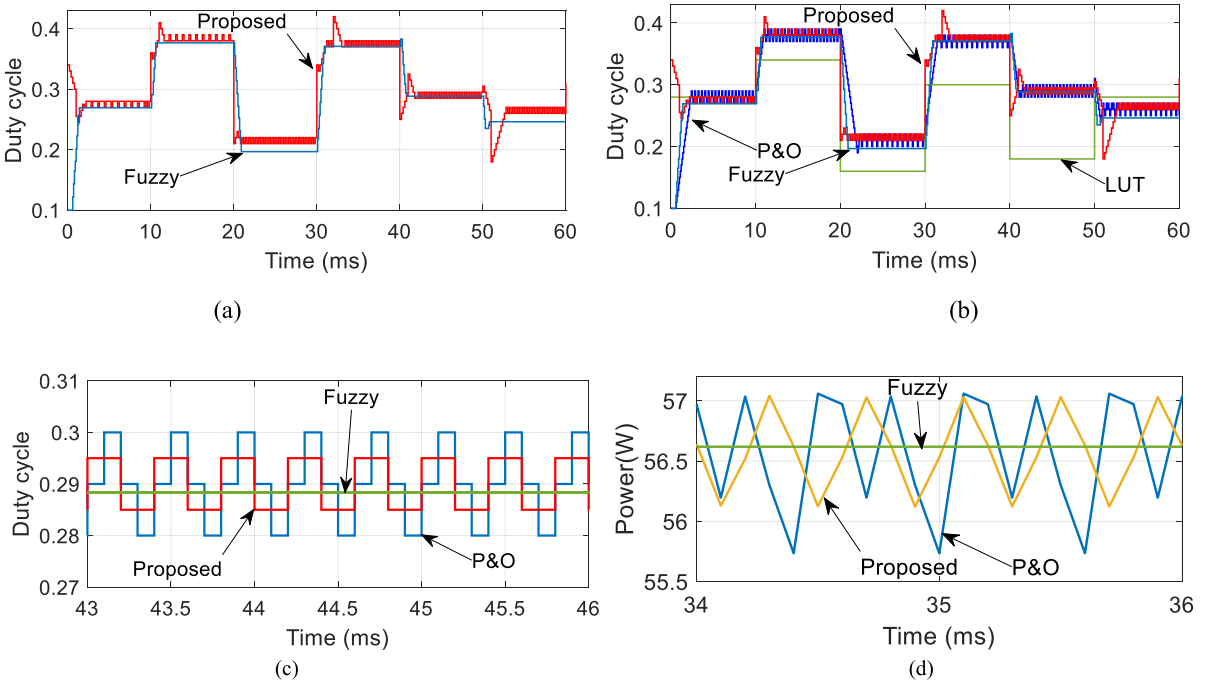


Fig. 4. (a) Duty cycle of the Two-stage Hybrid MPPT for step-changing inputs; (b) Duty cycle of the selected algorithms for step-changing inputs; (c) Duty cycle oscillation at the MPP; (d) Power ripple.

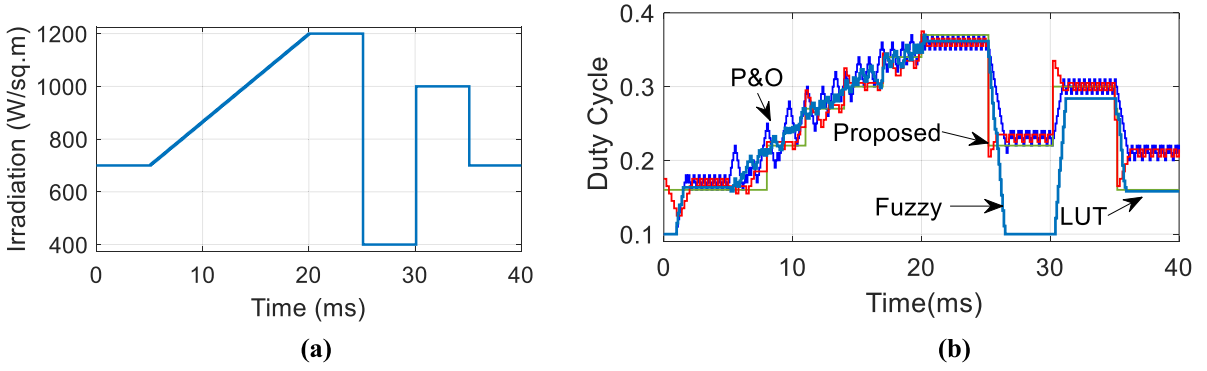


Fig. 5. (a) Rapidly varying irradiation, (b) Duty cycle output, of the selected and proposed MPPT for rapidly varying irradiation.

Therefore, the proposed MPPT technique is deemed appropriate for situations with swiftly changing circumstances. Table 7 presents the power ripple observed under different input conditions. The MPPT algorithm with fuzzy logic implementation exhibits no oscillation when operating at MPP. Therefore, this study solely examines the ripple analysis of conventional P&O for the purpose of comparison. The system employing the proposed MPP tracking technique exhibits a relatively lower level of output power ripple. The hybrid approach proposed in this study has been found to achieve a ripple reduction of around 30% while comparing it to the classical P&O algorithm.

5. Experimental validation

Fig. 6 depicts the hardware prototype of the photovoltaic (PV) system. The hardware undergoes testing in both controlled laboratory settings and real-world scenarios. The control circuits are powered by 30 V, 2 A power supply, and 30 V, 5 A regulated power supply is used as a source to test the hardware under laboratory environment. A 50 W PV module is used as a source to conduct the tests in a real environment. The key components of the hardware

Table 7. Comparison of output power ripple.

MPPT Methods	G in W/m^2 T in °C	600 25	1260 30	750 25	1000 25	965 38
Conventional P&O (%)		1.982	2.293	2.662	2.578	1.865
Proposed Hybrid (%)		1.641	1.693	1.917	1.910	1.542

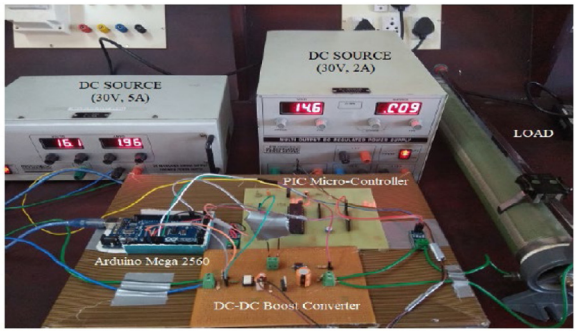


Fig. 6. Hardware setup of the proposed system.

circuit are the switch, diode, and the controller. IRFP460 A MOSFET is used for switch and 15ETH06 diode is used. The Arduino Mega 2560 microcontroller incorporates the MPPT algorithm. The acquisition of data involves the utilization of both the ACS712 current sensor and the voltage sensor. The empirical data, encompassing the duty cycle, input current, voltage, and output current, voltage are obtained and documented. The collected results are then imported to MATLAB, and plots are generated in MATLAB. The DC–DC converter receives input from the regulated power supply and a loading rheostat connected across the output of the converter as load. The hardware is tested with the proposed method and with P&O, LUT technique. Fig. 7(a) depicts the fluctuation in voltage that was applied to the converter to evaluate the P & O MPPT tracking. P&O MPPT duty cycle is depicted in Fig. 7(b). The controller monitors variations in the input(s) and the duty cycle is adjusted correspondingly to attain the MPP.

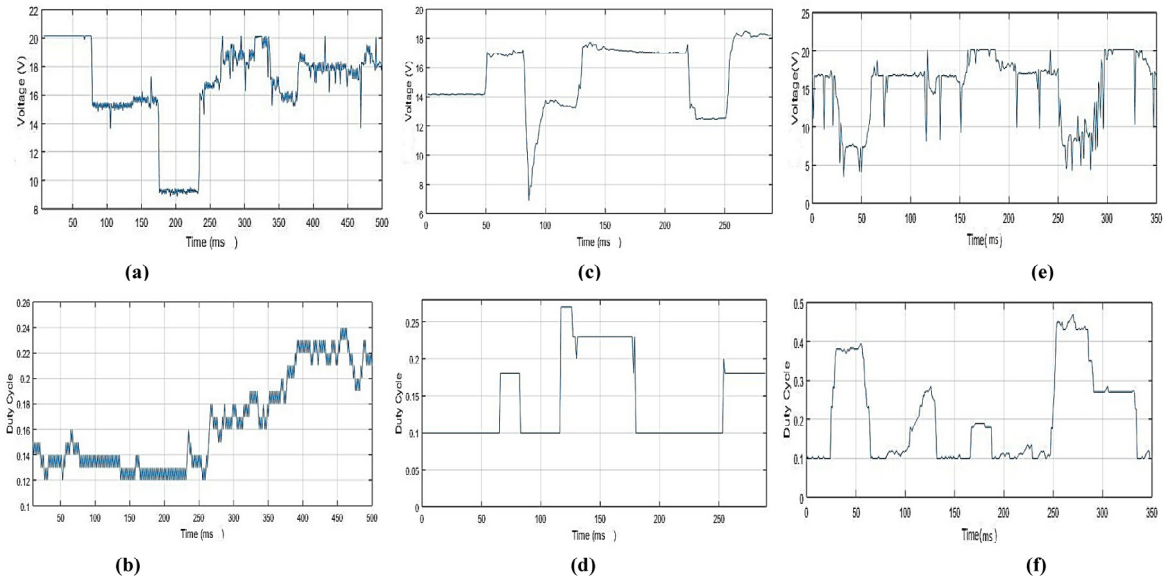
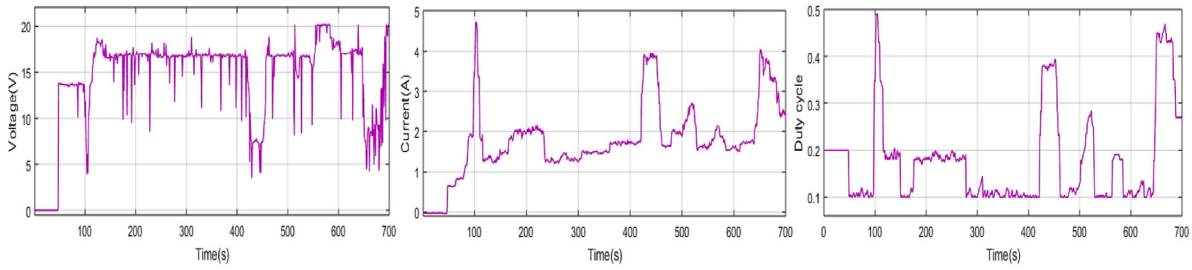


Fig. 7. Hardware results.



(a) (b) (c)

Fig. 8. (a) PV voltage; (b) PV current; (c) MPP tracking of proposed algorithm.

The LUT MPPT based system is also analyzed at laboratory conditions, and the results are obtained. Fig. 7(c) represents the input voltage to the converter with LUT MPPT. As depicted in Fig. 7(d), LUT MPPT provide a fixed duty cycle value, and the response of LUT MPPT corresponding to the input changes is faster than the P&O MPPT. The voltage applied to the PV converter with proposed two stage MPP tracking controller is presented in Fig. 7(e). LUT instantaneously provides the duty cycle at instances when large changes are detected by the controller. To achieve the MPP, the P&O algorithm also monitors and modifies the duty cycle. The proposed MPP tracking controller's duty cycle response is depicted in Fig. 7(f).

Experimental tests were conducted outdoors utilizing a 50 W solar PV panel as the source. Real-time analysis of the system was conducted between 11.30 am and 1.30 pm on a specific day, and the findings were documented. During real-time testing, shading sheets were used to fully shade the PV module. Fig. 8(a) shows the PV module output voltage, and the current is illustrated in Fig. 8(b). The duty cycle response of the proposed hybrid MPPT is depicted in Fig. 8(c). The suggested algorithm promptly adjusts to changes that occur in the output of PV voltage/current. The algorithm switches to LUT whenever a large variation is detected. Then, switching from LUT to P&O MPPT also occurs effectively, and the MPP is accurately tracked by the proposed MPPT.

Table 8 presents a quantitative comparison between some of the existing MPPT algorithms and the proposed one. The efficiency of a Two-level hybrid MPPT algorithm, which integrates Ripple Correlation Control (RCC) and a high frequency learning-based adjustable gain MRAC controller, has been reported to be 97.55% under different environmental conditions in a previous study [22]. The efficiency is comparatively low. Moreover, the approach is extremely complex, and the hardware requirement is high. Likewise, another hybrid MPPT algorithm proposed in Banakhr and Mosaad [23] which combines PI controller with harmony search optimization (PI-HS) is also a complex algorithm with lower tracking efficiency. This algorithm also uses a look up table approach where K_P and K_I values are stored. Finding optimal K_P and K_I values is a tedious process. This algorithm also needs high performance hardware for implementation. The proposed approach exhibits an overall efficiency of 99.31%. While the proposed approach exhibits a higher level of steady state oscillation due to its three-level swing at the MPP, it still provides greater efficiency compared to the drift-free MPPT approach. The proposed algorithm's tracking speed, complexity, and implementation cost were deemed satisfactory, alongside other significant factors. Based on the findings and analysis presented, it can be inferred that the hybrid MPPT method proposed in this article is appropriate for rapidly varying conditions.

Table 8. Comparison of various online, offline, and hybrid MPPT algorithms with the proposed algorithm.

MPPT	Type	Overall Efficiency	Tracking Speed	Implementation Complexity	Hardware Cost
VSS P&O [8]	Online	97.74%	Medium	Low	Low
ANN-PSO [12]	Hybrid	97.84%	High	High	Medium
FLC-EHO [13]	Hybrid	99.95%	High	High	Medium
Drift Free P&O [21]	Online	99.8%	Medium	Low	Low
HFLAG-MRAC [22]	Hybrid	97.55%	Medium	High	High
PI-HS [23]	Hybrid	91%–93%	High	High	High
Proposed LUT-P&O	Hybrid	99.31%	High	Low	Low

6. Conclusions

This paper presents a proposal for a self-contained solar photovoltaic (PV) system that utilizes a novel two-stage hybrid maximum power point tracking (MPPT) algorithm. The proposed approach integrates P&O LUT MPPT methods. The efficacy of the proposed method is assessed through both simulation-based investigations and practical implementation. The present study involves the simulation of a solar PV system utilizing multiple MPPT algorithms, including P&O, Fuzzy, LUT, and the proposed hybrid method. The system is evaluated under diverse input conditions, and the resulting simulation outcomes are analyzed. The performance analysis of the proposed MPPT is conducted with respect to its tracking speed and accuracy, oscillation at MPP, and output power ripple. Moreover, the efficacy of the proposed method is substantiated through hardware implementation. The experimental evaluation of the SPV system's hardware, incorporating P&O, LUT, and the proposed approach, was conducted in a laboratory setting, and the findings were subjected to analysis. The proposed MPPT algorithm has resulted in an overall tracking efficiency of 99.31%. The study concludes that the two-stage hybrid MPPT based PV system exhibits superior responsiveness compared to the other techniques evaluated, across a range of environmental conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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